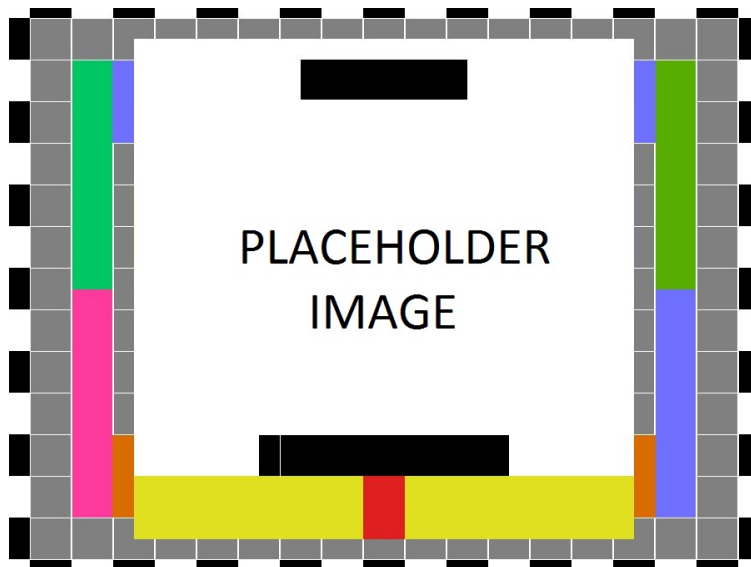


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM PERCEPTRON
DIGITDESK HANDWRITTEN FORM DIGITIZATION
SERVICE**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	09.02.2026	GH	document creation
1.0	09.15.2026	AT, GH, CB	official release

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1 PROBLEM STATEMENT

A county records office receives thousands of paper forms each month containing handwritten numeric fields (meter readings, account numbers, dated amounts). Staff key these into a database by hand — slow, costly, and error-prone, with mistakes surfacing only later as billing disputes. The office needs a faster, more consistent way to turn handwritten numeric fields on scanned forms into verified digital data.

2 METHODOLOGY

Team Perceptron will build DigitDesk Handwritten Form Digitization Service, a service that ingests scanned form images, locates the numeric fields, recognizes the handwritten digits with a machine-learning model, and returns structured, confidence-scored data. Low-confidence fields are routed to a human-in-the-loop review screen rather than accepted blindly. In short: we are building software that reads handwritten numbers off scanned forms and hands back clean data plus a short review queue.

3 VALUE PROPOSITION

For the sponsor, DigitDesk cuts manual keying effort and transcription errors while keeping a human check where the model is unsure, lowering downstream dispute costs. For the university, the project exercises a complete, responsible ML pipeline — data handling, training, evaluation, calibration, and human oversight — with a public benchmark (MNIST) available for early development.

4 DEVELOPMENT MILESTONES

- Project Charter first draft - September 2026
- System Requirements Specification - October 2026
- Architectural Design Specification - November 2026
- Demonstration of digit classifier on MNIST - December 2026
- Detailed Design Specification - February 2027
- Demonstration of field segmentation on real forms - March 2027
- Demonstration of review workflow & confidence routing - April 2027
- CoE Innovation Day poster presentation - April 2027
- Final Project Demonstration (end-to-end digitization) - May 2027

5 BACKGROUND

The office currently double-keys critical fields to catch errors, doubling labor. They approached the CSE program for an ML-assisted alternative that keeps a human in the loop for auditability. No prior relationship; requirements gathered via two interviews and a sample of redacted forms.

6 RELATED WORK

Handwritten-digit recognition is a classic ML benchmark (MNIST) [1], and commercial OCR/IDP products exist but are priced per-page and often cloud-only, which conflicts with the office's data-handling rules. DigitDesk targets an on-premises, auditable, confidence-routed pipeline. (Add ≥ 5 IEEE-format references in the final version.)

7 SYSTEM OVERVIEW

Scanned images enter a preprocessing stage (deskew, denoise, field segmentation); segmented digit crops go to the recognition model, which emits digits with confidence; results above a threshold are stored, and the rest enter a review queue served through a small web UI. See the ADS for the layered design.

8 PROJECT MANAGEMENT APPROACH

Iterative Scrum [1, 2] per ISO/IEC/IEEE 16326:2019 [3] within ISO/IEC/IEEE 12207:2017 [4]. Two-week sprints; estimation in task-hours; progress by burndown and by a tracked model-quality metric (field-level accuracy). Data and model versions are tracked alongside code in Git.

9 ROLES & RESPONSIBILITIES

Sponsor contact: records-office supervisor. Product Owner: Grace Hopper. Scrum Master: Alan Turing. Data/preprocessing: John Von Neumann; Model/eval: Ada Lovelace; API/storage: Charles Babbage; Review UI: Grace Hopper; Integration/test: Alan Turing.

10 COST PROPOSAL

Primarily cloud/GPU compute for training and an on-prem inference server; no custom hardware. Funded by the default CSE department stipend. Largest line item is GPU training hours.

11 FACILITIES & EQUIPMENT

Senior Design lab workstations, department GPU allocation for training, and an on-prem test server matching the sponsor environment. A sample of redacted real forms is required by the 3rd sprint.

12 ASSUMPTIONS

- Redacted real-form samples will be available by the 3rd sprint.
- Scans are provided at ≥ 200 dpi grayscale.
- Field locations are consistent across a given form template.
- Public MNIST data may be used for early model development.
- The sponsor can label a validation set of real fields.

13 CONSTRAINTS

- Final demonstration must be completed by May 1st, 2027.
- Total development cost must not exceed \$800.
- All form data must remain on sponsor-approved (on-prem) infrastructure.
- Personally identifiable fields must be redacted before any team access.
- Any accepted-without-review field must meet the agreed accuracy target.

14 RISKS

Risk description	Probability	Loss (days)	Exposure (days)
Real handwriting harder than MNIST (domain gap)	0.55	15	8.3
Field-segmentation fails on skewed/noisy scans	0.40	12	4.8
Insufficient labeled real data	0.45	10	4.5
On-prem GPU/server access delayed	0.25	12	3.0
Confidence calibration poor (bad routing)	0.30	8	2.4

Table 2: Highest-exposure project risks

15 DOCUMENTATION & REPORTING

Documentation deliverables tracked in Git and released at their planning sprints; recurring sprint artifacts and engineering notebooks maintained each sprint. Closeout: the deployed service, model card, poster, web page, demo video, documented source (MIT), deployment scripts, and an operator/reviewer manual.

REFERENCES

- [1] K. S. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*, 1st ed. Addison-Wesley Professional, 2012.
- [2] K. Schwaber and J. Sutherland, “The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game,” 2020.
- [3] ISO/IEC/IEEE, “ISO/IEC/IEEE 16326:2019 — Systems and software engineering — Life cycle processes — Project management,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2019.
- [4] —, “ISO/IEC/IEEE 12207:2017 — Systems and software engineering — Software life cycle processes,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2017.